



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
19/387,098

RECEIPT DATE / TIME
11/12/2025 04:29:19 PM Z ET

ATTORNEY DOCKET #
0737-000003-US

Title of Invention

SYSTEMS AND METHODS FOR MANAGING AN EVENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
 under 35 USC 111(a)

PATENT # -

CONFIRMATION # 9358

FILED BY ANGELA VOAKES

PATENT CENTER # 73154730

FILING DATE -

CUSTOMER # 110407

FIRST NAMED
INVENTOR Tiwanna Hayes

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY George Patsarikas

Documents

TOTAL DOCUMENTS: 5

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
0737-000003US_ADS.pdf	8	Application Data Sheet	2174 KB
0737-000003US Application-APP.TEXT.docx	36	Application body structured text document	45 KB
Warning: A "Drawings" section header has been detected in the specification. Please review, and revise if necessary.			
0737-000003-US Combined Assignment and Declaration_signed.pdf	1	Oath or Declaration filed	41 KB
0737-000003US Drawings.pdf	3	Drawings-only black and white line drawings	1591 KB
0737-000003US Application.pdf	36	Auxiliary PDF of Application	165 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
0737-000003US_ADS.pdf	F03351A276BFE6814D059A9E9D5DAA3A7F20E8B6FCE96394B 13B1E008A7670CB150A90DE1CCF135FB02F113188C9DB030D BE961646EBB91A9DF5F91354AFAD99
0737-000003US Application- APP.TEXT.docx	C4D38E062E7A912ACBCE438DAD543CAFE115425C88A428F6 1C1083A4920F6251093D345BF6CE5A44E214CC83869E865F45 36CE191F6DBB434CEE6C712F34DB8
0737-000003-US Combined Assignment and Declaration_signed.pdf	495D00D073068E7BEB9EE2C5F9081039817B6A85AAB527308 15F8A54619C996770419B89BDE44012E0639483A84A3B758B8 70BC18165F00FDE3DDB440C5C32FA
0737-000003US Drawings.pdf	02FBA5823A7C775575D2D628AF25FA63A0AB1880EDE3E278F 3B0CEC62FEFD76F2F8C010E0835DB37BF9D9980B41E0F1AF 93D5F300C04877B4649EC782E74E0D8
0737-000003US Application.pdf	30CFFD86AF3A243DA8EE3F1E8D34E0F49BC5B91EC7246975 FF12EFAEDE5D61FB75F433B7F299637C3C377B03AE395C7E 1DC4C723BFD72A1FB19E9953588EC5ED

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

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If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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PATENT AND TRADEMARK OFFICE

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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/387,098	11/12/2025 04:29:19 PM Z ET	0737-000003-US

Title of Invention

SYSTEMS AND METHODS FOR MANAGING AN EVENT

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	9358	FILED BY	ANGELA VOAKES
PATENT CENTER #	73154730	AUTHORIZED BY	George Patsarikas
CUSTOMER #	110407	FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Tiwanha Hayes

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 3112	E2025ABG29438051	ANGELA VOAKES

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
4011	BASIC FILING FEE- UTILITY	70.00	1	70.00
2111	UTILITY PATENT APPL. SEARCH FEE	308.00	1	308.00
2311	EXAMINATION OF ORIGINAL PATENT APPLICATION	352.00	1	352.00
TOTAL AMOUNT:				\$730.00

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement

Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0737-000003-US
		Application Number	
Title of Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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Inventor Information:

Inventor 1 Remove				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Tiwanne		Hayes	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Suwanee	Country of Residence ⁱ	US	
Mailing Address of Inventor:				
Address 1		3651 Peachtree Pkwy		
Address 2		STE E-339		
City	Suwanee	State/Province	GA	
Postal Code	30024	Country ⁱ	US	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence Information of this application.	
Customer Number	110407
Email Address	docketing@burrisiplaw.com Add Email Remove Email

Application Information:

Title of the Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		
Attorney Docket Number	0737-000003-US	Small Entity Status Claimed	☒
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	3	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0737-000003-US
		Application Number	
Title of Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	110407		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Claims benefit of provisional	63/720057	2024-11-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0737-000003-US
		Application Number	
Title of Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)

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Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0737-000003-US
		Application Number	
Title of Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0737-000003-US
		Application Number	
Title of Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1	Remove		
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117		
☐ Person to whom the inventor is obligated to assign.	☐ Joint Inventor		
☐ Person who shows sufficient proprietary interest			
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
Organization Name	Resilient Reach IP Holdings, LLC		
Mailing Address Information For Applicant:			
Address 1	3651 Peachtree Pkwy		
Address 2	STE E-339		
City	Suwanee	State/Province	GA
Country	US	Postal Code	30024
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0737-000003-US
		Application Number	
Title of Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

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If the Assignee or Non-Applicant Assignee is an Organization check here.



Organization Name

Resilient Reach IP Holdings, LLC

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1		3651 Peachtree Pkwy	
Address 2		STE E-339	
City		Suwanee	State/Province GA
Country ⁱ	US	Postal Code	30024
Phone Number		Fax Number	
Email Address			

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

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NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is **not** checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/George A. Patsarikas/		Date (YYYY-MM-DD)	2025-11-12
First Name	George A.	Last Name	Patsarikas	Registration Number 74,670

Additional Signature may be generated within this form by selecting the Add button.

[Add](#)

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0737-000003-US
		Application Number	
Title of Invention	SYSTEMS AND METHODS FOR MANAGING AN EVENT		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

SYSTEMS AND METHODS FOR MANAGING AN EVENT

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to U.S. provisional application number 63/720,057 filed on November 13, 2024. The disclosure of the above application is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to the management of an event, and more particularly, to systems and methods for predicting an outcome of one or more scenarios associated with the event.

BACKGROUND

[0003] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0004] Event management software currently focuses primarily on event planning, ticketing, and marketing. However, existing software does not consider potential disruptions that may occur in management of the event.

SUMMARY

[0005] This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

[0006] In some embodiments, the techniques described herein relate to a method for managing an event, the method including: obtaining a dependency graph including a plurality of nodes representing operational elements of the event and a

plurality of edges representing relationships between the operational elements; generating, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine; and performing a graph updating routine, wherein the graph updating routine includes: applying a modification to a first set of nodes of the plurality of nodes; identifying a subgraph of the dependency graph based on the modification, wherein the subgraph includes a second set of nodes of the plurality of nodes; generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector; and displaying the modified dependency graph.

[0007] In one or more variations of the method of the above paragraph, which may be implemented alone or in any combination: the graph updating routine includes a scenario simulation routine or a bounded propagation routine; applying the modification further includes modifying, for each node of the first set of nodes, at least one risk score of the risk vector; the at least one risk score includes a likelihood value, an impact value, a detection confidence value, and a data quality value; the modification includes modifying, for each node of the first set of nodes, at least one readiness score of the readiness vector; the at least one readiness score includes at least one of a completion status value, an inbound risk propagation values, a service level agreement (SLA) impact factor value, and a reliance value; the analytic routine includes at least one of a graph-based analytic routine, a statistical analytic routine, a supervised learning routine, a sequence-based routine, and an unsupervised learning routine; the method further includes selectively adjusting, for a first node of the plurality nodes, at least one of the risk vector and the readiness vector when a relationship type associated with the first node is a Blocks type; a number of nodes of the second set of nodes is based on at least

one of a convergence threshold and a maximum iteration count; and/or the method further includes terminating the graph updating routine when (i) at least one readiness score of the readiness vector is less than the convergence threshold or (ii) an iteration value of the graph updating routine is equal to the maximum iteration count.

[0008] In some embodiments, the techniques described herein relate to a system for managing an event, the system including: a processor; and a nontransitory computer-readable medium including instructions that are executable by the processor, wherein the instructions include: obtaining a dependency graph including a plurality of nodes representing operational elements of the event and a plurality of edges representing relationships between the operational elements; generating, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine; and performing a graph updating routine, wherein the graph updating routine includes: applying a modification to a first set of nodes of the plurality of nodes; identifying a subgraph of the dependency graph based on the modification, wherein the subgraph includes a second set of nodes of the plurality of nodes; generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector; and displaying the modified dependency graph.

[0009] In one or more variations of the method of the above paragraph, which may be implemented alone or in any combination: the graph updating routine includes a scenario simulation routine or a bounded propagation routine; applying the modification further includes modifying, for each node of the first set of nodes, at least one risk score of the risk vector; the at least one risk score includes a likelihood value, an impact value, a detection confidence value, and a data quality value; the modification

includes modifying, for each node of the first set of nodes, at least one readiness score of the readiness vector; the at least one readiness score includes at least one of a completion status value, an inbound risk propagation values, a service level agreement (SLA) impact factor value, and a reliance value; the analytic routine includes at least one of a graph-based analytic routine, a statistical analytic routine, a supervised learning routine, a sequence-based routine, and an unsupervised learning routine; the techniques described herein relate to a system, further including selectively adjusting, for a first node of the plurality nodes, at least one of the risk vector and the readiness vector when a relationship type associated with the first node is a Blocks type; a number of nodes of the second set of nodes is based on at least one of a convergence threshold and a maximum iteration count; and/or the instructions further include terminating the graph updating routine when (i) at least one readiness score of the readiness vector is less than the convergence threshold or (ii) an iteration value of the graph updating routine is equal to the maximum iteration count.

[0010] In some aspects, the techniques described herein relate to a system for managing an event, the system including: a processor; and a nontransitory computer-readable medium including instructions that are executable by the processor, wherein the instructions include: generating a dependency graph including a plurality of nodes representing operational elements of the event and a plurality of edges representing relationships between the operational elements; generating, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine, wherein the analytic routine includes at least one of a graph-based analytic routine, a statistical analytic routine, a supervised learning routine, a sequence-based routine, and an

unsupervised learning routine; and performing a graph updating routine, wherein the graph updating routine includes a scenario simulation routine or a bounded propagation routine, and wherein the graph updating routine includes: applying a modification to a first set of nodes of the plurality of nodes; identifying a subgraph of the dependency graph based on the modification, wherein the subgraph includes a second set of nodes of the plurality of nodes; generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector; and displaying the modified dependency graph. In some embodiments, applying the modification further includes modifying, for each node of the first set of nodes, at least one risk score of the risk vector and at least one readiness score of the readiness vector.

[0011] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

DRAWINGS

[0012] In order that the disclosure may be well understood, there will now be described various forms thereof, given by way of example, reference being made to the accompanying drawings, in which:

[0013] FIG. 1 is a block diagram illustrating an example computer system in accordance with one or more embodiments of the present disclosure;

[0014] FIG. 2 is a block diagram of an event management system in accordance with one or more embodiments of the present disclosure;

[0015] FIG. 3 schematically illustrates a dependency graph in accordance with one or more embodiments of the present disclosure; and

[0016] FIG. 4 is a flowchart illustrating an event management method in accordance with one or more embodiments of the present disclosure.

[0017] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

DETAILED DESCRIPTION

[0018] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

[0019] The present disclosure provides systems and methods for predicting outcomes of one or more scenarios associated with an event. Such predictions improve operational efficiency, communication, cybersecurity, and crisis management for organizations managing complex or large-scale operations and events. The artificial intelligence (AI)-driven platform described herein enables proactive management of these operations by providing real-time insights and scenario simulations that enhance resource allocation and decision-making.

[0020] In some embodiments, the event management system offers intelligent dependency mapping that provides real-time visualization of interrelated tasks and dependencies, facilitating proactive management and bottleneck identification. Using a graph-based dependency modeling approach, the event management system represents operational relationships and dependencies, enabling continuous assessment

and updates without requiring full recalculation of the dependency network. The event management system may also include impact modeling features that employ predictive analytics to simulate outcomes and enhance decision-making across multiple scenarios. Through proactive cyberthreat defense capabilities, the platform detects and mitigates potential threats in real time, enhancing operational security.

[0021] In some embodiments, the event management system may also include a comprehensive graphical user interface (GUI) that dynamically presents and updates operational data, improving both user efficiency and interaction performance. The GUI enables guided input of data sets and displays predictive results in an intuitive format, thereby improving human–machine interaction and overall usability.

[0022] In some embodiments, the event management system facilitates simulation and evaluation of possible scenarios based on received data sets. The system maps interdependencies among data sets, assigns risk and readiness scores to each operational pathway, and determines optimal responses. Through these simulations, users can view and select specific scenarios that improve decision-making by providing predictive insights into potential outcomes. In some embodiments, the event management system may automatically process multiple data sets from various sources related to an event, thereby increasing the likelihood of generating desirable and realistic scenarios. Processing a greater number of data sets enhances the system’s decision-making accuracy and operational readiness, while reducing computational overhead through efficient, bounded propagation techniques. That is, unlike conventional event management systems that require global recomputation, the disclosed system improves

computing efficiency by performing bounded recomputation and selective subgraph analysis, reducing the computational complexity of real-time updates.

[0023] In some embodiments, the event management system performs real-time learning routines to continuously adapt and improve performance based on incoming operational data and feedback. These routines allow for dynamic adjustment of predictions and readiness assessments, improving resilience and accuracy in event management operations.

[0024] The event management system may further implement orchestration routines that coordinate between operational elements, assessing task dependencies, resource availability, and timeline constraints to generate predictive readiness insights. This orchestration enables proactive identification of potential issues and supports timely intervention to maintain operational objectives.

[0025] The embodiments described herein provide an integrated, AI-driven event management platform that delivers real-time, actionable insights, enhances operational efficiency, and improves the underlying computing functionality. By combining dependency mapping, predictive modeling, cybersecurity, and orchestration capabilities, the event management system provides organizations with a tailored, adaptive framework that anticipates and mitigates operational risks.

[0026] FIG. 1 illustrates an operating environment that facilitates the performance of the one or more systems and methods described herein. In some embodiments, the systems and methods described herein can be implemented using the computing device 102. For example, the computing device 102 can be a personal computer, a desktop, a laptop, a tablet, a hand-held computer, a server, a workstation, a

mainframe, a wearable computer, a supercomputer, or a combination thereof. However, it is understood that the above-referenced examples of the computing device 102 is non-exhaustive and the computing device 102 can be any type of processing or computing device.

[0027] In some embodiments, the computing device 102 may include a processor 104, a display adapter 106, one or more input/output port(s) 108, one or more input/output component(s) 110, a network adapter 112, a power supply 114, and a memory 116. However, it is understood that the computing device 102 can include any additional components therein and is not required to include any of the listed components (e.g., the processor 104, the display adapter 106, the one or more input/output port(s) 108, the one or more input/output component(s) 110, the network adapter 112, the power supply 114, and the memory 116).

[0028] In some embodiments, the processor 104 is configured to provide instructions to the computing device 102 so that the computing device 102 can process one or more tasks including the implementation of a software program to perform one or more operations as described in more detail herein. It is also understood that the computing device 102 may include any number of processors 104 therein. For example, the processor 104 may be integrated with any other existing systems (e.g., ERP, CRM, project management tools, etc.) to utilize any data elements as described herein. As another example, the processor 104 can perform and/or facilitate performance of one or more security measures including end-to-end encryption for any and all transmission of data; multi-factor authentication for platform access; role-based access control that provides only authorized users can access sensitive data; or a combination thereof.

[0029] In some embodiments, the display adapter 106 can be a graphics card or a video board that provides the computing device 102 with a capability to display content on a display device 118. For example, the display device 118 can be any screen, monitor, and/or light-emitting component associated with any of the personal computer, the desktop, the laptop, the tablet, the hand-held computer, the server, the workstation, the mainframe, the wearable computer, the supercomputer, or a combination thereof. However, it is understood that the above-referenced examples of the display device 118 is non-exhaustive, and the display device 118 can be any type of device capable of providing a visual display.

[0030] In some embodiments, the input/output port(s) 108 provide interfaces (e.g., sockets) for one or more cables to connect to the computing device 102. It is understood that there may be any number of input/output port(s) 108 on the computing device 102. For example, the input/output port(s) 108 enables the reception of signals and/or data from an external device connected to the computing device 102 via the one or more cables. As another example, the input/output port(s) 108 enables the transmission of signals and/or data to an external device connected to the computing device 102 via the one or more cables. The input/output component(s) 110 can include one or more components that support the input/output port(s) 108 such as, but not limited to, a switch, a push button, a pressure mat, a float switch, a keypad, a radio receive, or a combination thereof.

[0031] The network adapter 112 can be any type of network interface controller that enables communication over a network 120 with another computing device, such as a remote computing device 122. For example, the network 120 may be a secure

and scalable cloud infrastructure capable of hosting a platform for providing flexible and accessible management of an event, as is described herein. As an example, the remote computing device 122 can be a user device, such as a cellular-phone, a smartphone, a tablet, a laptop, or a combination thereof. The power supply 114 is configured to convert alternating high voltage current (e.g., AC) into direct current (e.g., DC) to provide regulated power to the other components (e.g., the processor 104, the display adapter 106, the one or more input/output port(s) 108, the one or more input/output component(s) 110, the network adapter 112, and the memory 116) of the computing device 102.

[0032] Additionally, the memory 116 can be a mass storage device and/or a system memory such as a hard disk drive, a memory card, a solid-state drive, random access memory (RAM), or a combination thereof. The memory 116 is configured to store instructions and data associated with the operation of the computing device 102. For example, the data associated with the operation of the computing device 102 can include operational data, dependency data, risk data, readiness data, impact data cybersecurity data, or a combination thereof. As another example, the operational data can include task details, timeliness, resource allocation, or a combination thereof. As yet another example, the dependency data can include interconnections between tasks, teams, resources, or a combination thereof. As a further example, the risk data can include historical data, predictive analytical outputs, risk scores, or a combination thereof. For example, the readiness data can include task completion percentages, readiness scores, or a combination thereof. As an additional example, the impact data can include scenario outcomes, potential downstream effects, or a combination thereof. As a further example, the cybersecurity data can include threat detection logs, encrypted communications, or a

combination thereof. The memory 116 can generally include an operating system 124, modeling software 126, and modeling data 128. For example, the operating system 124 is configured to manage and/or process any of the data and/or instructions associated with the modeling software 126 and/or modeling data 128, as described in more detail herein.

[0033] Furthermore, a system bus 130 is also included within the computing device 102 that is configured to couple each of the various components (e.g., the processor 104, the display adapter 106, the one or more input/output port(s) 108, the one or more input/output component(s) 110, the network adapter 112, the power supply 114, and the memory 116) of the computing device 102. It is also understood that each of the components of the computing device 102, and the functionality associated with each of the components of the computing device 102, may be implemented within the remote computing device 122. While the operating environment illustrated within FIG. 1 depicts a particular configuration associated with at least the computing device 102, the network 120, and the remote computing device 122, it is understood that the operating environment may be configured in any way.

[0034] In some embodiments, the computing device 102 is configured to identify, assess, and/or mitigate any operational risks associated with managing a workflow associated with an event. It is understood that the event can be any planned occasion such as, but not limited to, a sporting event, large gathering, or a musical event. It is also understood that the event can be any unplanned occasion. As such, the various embodiments described herein can be used in connection with any type of event or gathering (formal or otherwise). It is also understood that the workflow can be, but is not

limited to, a business process associated with the event, a security process associated with the event, among others.

[0035] For example, and referring to FIG. 2, the computing device 102 is implemented within an example event management system 200, which includes an ingestion module 205, a mapping module 210, a risk-readiness module 220, a score adjustment module 225, a scenario simulation module 240, a propagation module 250, and a user interface 260.

[0036] In some embodiments, the ingestion module 205 is configured to obtain one or more data sets. For example, the one or more data sets is obtained as part of an onboarding process. As another example, the one or more data sets can be received from one or more sources including, but not limited to, a governing organization associated with the event and/or any vendors associated with the event. As yet another example, the one or more data sets can include organizational data, one or more training manuals, organization chart(s), organizational operating procedures, one or more tasks, one or more dependencies, one or more risks, one or more resources, or a combination thereof. It is understood, however, that the one or more data sets can include any data elements associated with the event.

[0037] In some embodiments, the mapping module 210 is configured to model operational dependencies and relationships of the one or more data sets. In some embodiments, and with reference to FIGS. 2-3, the mapping module 210 represents operations as a dependency graph 270, where nodes N represent individual tasks, activities, or operational elements, and edges E represent relationships between these elements. The dependency graph 270 provides a computational framework for tracking

relationships between operational tasks, resources, and events within complex event management scenarios.

[0038] In some embodiments, the dependency graph 270 utilizes typed edges that define specific relationship categories between nodes. The typed edges may include, but are not limited to, Precedes, Enables, Blocks, Supplies, and Escalates relationships. Precedes relationships may establish temporal ordering between operational tasks. Enables relationships may define conditional dependencies where one node must be completed or activated before another node can proceed. Supplies relationships may indicate resource provision dependencies where one node provides resources, data, or services to another node. Escalates relationships define hierarchical dependencies where issues or status changes in one node trigger elevated priority or attention in connected nodes. Blocks relationships may represent inhibitory connections where one node prevents or delays the execution of another node.

[0039] In some embodiments, each edge in the dependency graph 270 is configured with multiple attribute parameters that quantify the relationship characteristics, such as criticality levels, temporal slack values, reliance factors, service level agreement (SLA) impact measurements, and/or confidence scores. The criticality attribute measures the importance of the dependency relationship on a numerical scale, indicating how failure or delay in the relationship affects overall system performance. The temporal slack attribute represents the available time buffer between connected nodes and may be measured in time units, such as minutes or hours. The reliance attribute quantifies the degree of dependence between nodes, indicating how strongly one node depends on another for successful completion. The SLA impact attribute measures the effect of the

dependency relationship on SLA compliance and may be expressed as a numerical impact score. The confidence attribute represents the reliability or certainty of the dependency relationship and may be expressed as a probability value between 0 and 1.

[0040] In some embodiments, nodes within the dependency graph maintain multiple attribute categories that track operational state and performance metrics. The status attribute records the current operational state of each node, such as pending, active, completed, or failed states. The completion attribute tracks the percentage or degree of task completion for each node and may be represented as a value between 0 and 100 percent. The signals attribute stores real-time data inputs, alerts, or notifications associated with each node, which are used for monitoring and decision-making processes. The risk scores attribute maintains calculated risk assessment values that quantify potential negative outcomes or failures associated with each node. The readiness scores attribute stores computed readiness metrics that indicate how prepared or capable each node is to execute its assigned operations.

[0041] Accordingly, the dependency graph 270 enables the system to model complex operational relationships by connecting nodes through the typed edge relationships while maintaining detailed attribute information at both the node and edge levels. The dependency graph 270 allows for dynamic updates to node attributes as operational conditions change, while the edge attributes provide context for how changes propagate through the dependency graph 270. The combination of typed edges and comprehensive attribute systems creates a detailed operational model that supports real-time monitoring, risk assessment, and predictive analysis for event management scenarios.

[0042] In some embodiments, the risk-readiness module 220 processes the node and edge data of the dependency graph 270 to generate quantitative assessments. As an example, the risk-readiness module 220 computes risk vectors for each node, where each risk vector comprises likelihood values, impact values, detection confidence values, and data quality values.

[0043] The likelihood value may represent the probability that a particular risk event will occur at a given node, and it may be expressed as a numerical value between 0 and 1, where higher values indicate greater probability of occurrence. For example, a node representing equipment setup for an outdoor event receives a higher likelihood score during periods of forecasted severe weather compared to clear weather conditions. The impact value may quantify the magnitude of consequences that would result if the risk event materializes, measures the potential disruption to downstream operations, measures resource requirements for remediation, and measures overall effect on event readiness. For example, a power system failure at a venue receives a higher impact score than a minor audio equipment malfunction due to the broader operational consequences. The detection confidence value may indicate the reliability and accuracy of the data sources used to assess the risk. This component accounts for sensor accuracy, data freshness, and the completeness of available information. For example, real-time sensor data from monitored equipment provides higher detection confidence compared to manually reported status updates that is obsolete. The data quality value may evaluate the integrity and consistency of the underlying information used in risk assessment, and it may consider factors such as data completeness, consistency across multiple sources, and the presence of anomalies or outliers. For

example, automated data feeds from integrated systems may provide higher data quality scores compared to manually entered information that is subject to human error.

[0044] In some embodiments, each value of the risk vector is weighted to reflect the relevance of that value within a specific operational context. The weighted combination process allows the system to adapt risk calculations to different operational contexts and priorities, as different event types, venues, and operational phases utilize different weighting factors to emphasize relevant risk components for each specific situation.

[0045] As an example, for a security checkpoint node during a large sporting event, the likelihood value receives a moderate score based on historical crowd flow patterns and current attendance projections. Furthermore, the impact value receives a high score because security delays affect multiple downstream operations including concession sales, seating, and emergency egress procedures. The detection confidence value may receive a high score due to real-time crowd monitoring sensors and automated turnstile data. The data quality value may receive a moderate score because some manual security reports supplement the automated data streams. Accordingly, the weighted combination of these values produces a risk score that reflects the overall risk profile of the security checkpoint operation.

[0046] As another example, for a catering preparation node, the likelihood value receives a low score when adequate preparation time remains, and all ingredients are confirmed available. Furthermore, the impact value receives a moderate score because catering delays affect guest satisfaction but do not prevent the event from proceeding. The detection confidence value may receive a high score due to integrated

inventory management systems and real-time preparation tracking, and the data quality component may receive a high score because automated systems provide consistent and complete information. Accordingly, the resulting risk vector reflects the relatively low overall risk associated with the catering operation under these conditions.

[0047] In some embodiments, the risk-readiness module 220 also generates readiness vectors for one or more of the nodes, where each of the readiness vectors includes one or more readiness scores. As an example, the readiness scores may be generated using monotonic functions (e.g., where higher completion levels and lower risk factors produce increased readiness values) that are based on completion status values, inbound risk propagation values, SLA impact factor values, and reliance values.

[0048] In some embodiments, the completion status value represents the current progress state of a task or operation within the dependency graph 270, and the completion status value may be a normalized value between zero and one, where zero indicates no progress and one represents full completion. In some embodiments, the inbound risk propagation value indicates a risk propagation from upstream dependencies within the dependency graph 270. As an example, the risk-readiness module 220 aggregates risk values from connected predecessor nodes and applies weighting factors based on the strength and type of dependency relationships. Higher inbound risk values reduce the readiness score for the target node, reflecting the potential impact of upstream issues on downstream operations.

[0049] In some embodiments, the SLA impact factor value indicates the potential consequences of delays or failures on contractual obligations and performance

targets. The risk-readiness module 220 assigns impact weights based on the severity of potential SLA violations, with operations having greater SLA exposure receiving higher impact scores. In some embodiments, the SLA impact factor value impacts inversely affects readiness scores, where higher SLA impact reduces the computed readiness value. In some embodiments, the reliance values quantify the degree of dependency between connected nodes in the dependency graph 270. The risk-readiness module 220 may determine the reliance value based on the number and strength of incoming dependencies, with nodes having multiple or strong dependencies receiving higher reliance scores. Increased reliance may reduce readiness scores, as highly dependent operations may face greater vulnerability to upstream disruptions. In some embodiments, each readiness score of the readiness vector is weighted to reflect the relevance of that value within a specific operational context in a similar manner as the risk scores of the risk vector.

[0050] To perform the functionality described herein, the risk-readiness module 220 may perform one or more analytic routines, such as graph-based analytic routines, statistical analytic routines, supervised learning routines, sequence-based routines, and/or unsupervised learning routines. The implementation of multiple routines may provide a comprehensive analytical framework where each method contributes specialized capabilities to the overall risk and readiness assessment process. As an example, the graph-based analytic routine may provide structural understanding, statistical analytic routines may offer predictive insights, supervised learning routines may enable pattern-based classification, sequence-based routines may capture temporal dynamics, and unsupervised routines may detect anomalous conditions.

[0051] In some embodiments, graph-based analytic routines may include propagation routines, path minima calculations routines, and centrality-informed bottleneck identification routines. As an example, the propagation routines may include traversing the dependency graph 270 while considering the specific characteristics of each node and edge relationship. As another example, path minima calculation routines may identify bottleneck points within dependency chains by analyzing the minimum capacity or readiness values along critical pathways. As another example, centrality-informed bottlenecking routines may include evaluating node importance based on graph centrality measures, allowing the risk-readiness module 220 to prioritize nodes that have the greatest influence on overall event readiness.

[0052] In some embodiments, statistical analytic routines may include linear-based and logistic-based relationship analysis between input variables and outcome probabilities, generating probability estimates for various risk scenarios. For example, the statistical analytic routines may include exponentially weighted moving average (EMWA) routines that track trending patterns in data streams by applying greater weight to recent observations while maintaining historical context. As another example, the statistical analytic routines may include statistical process control (SPC) drift detection routines that monitor data streams for statistically significant deviations from established baseline patterns, triggering alerts when process variations exceed control limits.

[0053] In some embodiments, the supervised learning routines enhance the event management system's 100 pattern recognition capabilities through trained classification and regression algorithms. As an example, the supervised learning routines may include tree-ensemble model routines, including random forest and gradient boosting

routines, which analyze feature interactions to predict risk levels and readiness states. These supervised classifiers may process multiple input features simultaneously to generate classification outputs that inform risk assessment decisions. The supervised learning models learn from labeled training examples to identify patterns that correlate with specific risk or readiness outcomes.

[0054] In some embodiments, the sequence-based routines may analyze temporal patterns within the data set to identify time-dependent relationships and trends. These models process sequential data inputs to detect patterns that emerge over time, such as recurring failure modes or predictable performance degradation cycles. The sequence analysis capabilities enable the system to anticipate future states based on observed temporal progressions in the data set. In some embodiments, the unsupervised learning routines may identify anomalous patterns or outliers without requiring pre-labeled training data. As an example, autoencoding routines may learn normal operational patterns by training neural networks to reconstruct input data, with reconstruction errors serving as anomaly indicators.

[0055] In some embodiments, the score adjustment module 225 is configured to selectively adjust the risk scores and/or readiness scores generated by the risk-readiness module 220 based on one or more rules. As an example, the one or more rules may include a blocking rule for dependencies having a “Blocks” relationship, which includes adjusting the aggregated value of the readiness vector to zero. For example, in a sports event scenario, a field preparation task may depend on equipment delivery and weather clearance. If equipment delivery is marked as a blocking dependency (i.e., a “Blocks” relationship) and remains incomplete, the score of the aggregated value of the

readiness vector is adjusted to zero despite favorable weather conditions and available personnel. As another example, the one or more rules may include a critical path penalty rule, which includes reducing the aggregate readiness score when critical path nodes experience low readiness values or blocking conditions exist. As an example, if a critical path node shows 60% readiness while non-critical nodes average 90% readiness, the penalty adjustment reduces the overall aggregate score below the simple weighted average to reflect the disproportionate impact of critical path delays. The penalty calculation applies multiplicative factors based on critical path node performance, where critical path readiness below threshold values triggers progressively larger penalty adjustments.

[0056] To perform the functionality described herein, the score adjustment module 225 is configured to employ rule-based evaluators, temporal gates, and/or exception dictionaries. Rule-based evaluators may implement SLA threshold evaluators that compare current performance metrics against predefined SLA parameters. Temporal gates may function as time-based decision points that evaluate whether specific conditions are met within designated time windows. Exception dictionaries may maintain collections of predefined scenarios and their corresponding response protocols, enabling identification and handling of known problematic conditions.

[0057] In some embodiments, the scenario simulation module 240 is configured to perform predictive analytic-based routines on the dependency graph 270 scenarios as modifications to nodes and edges of the dependency graph 270 within specified time windows, along with associated objectives and constraints. As an example, the scenario simulation module 240 may copy the dependency graph 270, applies the

defined modifications (e.g., increasing the completion status of a security checkpoint node from 60% to 100% within a two-hour time window), identifies affected subgraphs, and determines the risk and readiness scores for the impacted nodes of the affected subgraph.

[0058] Accordingly, the scenario simulation module 240 may evaluate multiple scenarios in parallel that identify trade-offs between different approaches (e.g., a high-cost scenario achieving 95% readiness against a moderate-cost scenario achieving 87% readiness) with improved computational efficiency. That is, multiple scenario simulations may run concurrently by creating independent copies of the dependency graph 270 and applying different modification sets to each copy. Moreover, by determining the risk vectors and readiness scores for nodes within the affected subgraphs as opposed to the entire dependency graph 270, the amount of computing resources (e.g., processor-based, resource-based, and/or time-based resources).

[0059] The propagation module 250 is configured to update the dependency graph 270 by performing a bounded propagation routine. In some embodiments, the bounded propagation routine employs a priority queue of downstream edges to manage the propagation of changes through the dependency graph 270. In some embodiments, the priority queue organizes edges based on propagation priority, ensuring that changes flow through the graph in an ordered manner. The bounded propagation routine may process changes by selecting edges from the priority queue based on their propagation weights, which may be determined by factors such as edge criticality, temporal urgency, and dependency strength. The bounded propagation routine may calculate updated risk and readiness vectors for target nodes by applying propagation functions that consider

the magnitude of change, edge attributes, and existing node states. The bounded propagation routine may be iteratively performed, with each iteration adding newly affected downstream edges to the priority queue until convergence criteria are satisfied.

[0060] In some embodiments, the propagation module 250 terminates propagation calculations when incremental changes fall below a predetermined convergence threshold ϵ is reached (i.e., indicating that further propagation would produce negligible effects) or when a maximum iteration count θ is reached (e.g., a computational bound regardless of convergence behavior). That is, the number of nodes in the subgraph may be based on at least one of the convergence threshold and the maximum iteration count, where the propagation routine continues adding downstream nodes to the affected set until either the convergence criteria is satisfied or the maximum iteration count θ is exceeded. Accordingly, the bounded propagation routine reduces the computational complexity from $O(N)$ for a full dependency graph update to $O(k \cdot d)$, where k represents the number of affected nodes and d represents the average node degree.

[0061] As an example, when the dependency graph 270 includes 10,000 nodes having 50 nodes are affected by a status update, and where each node has an average of 5 connections, the computational requirement reduces from 10,000 operations to approximately 250 operations to propagate the updates to the dependency graphs 270. As a more specific example, when a venue security checkpoint node experiences a status change from "delayed" to "operational," the bounded propagation routine may first update the immediately connected concession stand nodes, then propagate to seating management nodes, and finally to emergency egress nodes. In this scenario, if the security checkpoint change affects 15 downstream nodes (as indicated by the

convergence threshold and/or the maximum iteration count) with an average of 3 edge connections each, the bounded propagation processes approximately 45 edge updates rather than recalculating the entire 10,000-node dependency graph, thereby reducing computational overhead while maintaining accuracy in the affected operational areas.

[0062] In some embodiments, the user interface 260 is configured to provide real-time visualization and interaction capabilities. As an example, the user interface 260 may display the dependency graph 270, one or more scores of the risk vectors, and/or one or more scores of the readiness vectors. As another example, the user interface 260 receives updates from the risk-readiness module 220 to display current readiness states and automatically refresh displayed information as conditions change. As yet another example, the user interface 260 visualizes the dependency relationships from the mapping module 210, highlights paths identified as having high impact on overall readiness, and presents ranked mitigation options generated by the scenario simulation module 240. In some embodiments, the user interface 260 may include comparison views that display readiness changes, resource requirements, and implementation timeframes to support rapid decision-making processes. To perform the functionality described herein, the user interface 260 may include a display panel, one or more input devices such as a touchscreen, keyboard, or pointing device, and a processing module configured to receive user input signals, generate corresponding control commands, and present visual or graphical output on the display in response to user interactions.

[0063] Referring to FIG. 4, a flowchart of a method 400 for managing an event is shown. The method 400 begins at step 402, where the event management system 200 obtains a dependency graph comprising a plurality of nodes representing

operational elements of the event and a plurality of edges representing relationships between the operational elements (e.g., the dependency graph 270). At step 404, the event management system 200 generates, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine. At step 406, the event management system 200 performs a graph updating routine (e.g., a scenario simulation routine or a bounded propagation routine). In some embodiments, the graph updating routine may include applying a modification to a first set of nodes of the plurality of nodes and identifying a subgraph of the dependency graph based on the modification (e.g., a subgraph of the dependency graph 270). The graph updating routine may also include generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector and displaying, using the user interface 260, the modified dependency graph.

[0064] As described herein, the event management system 200 provides a comprehensive solution for managing complex events through dependency modeling, risk assessment, and predictive analytics. The system may utilize graph-based representations to model operational relationships and dependencies, enabling real-time monitoring and proactive management of event operations. Through the combination of multiple analytic routines including graph-based, statistical, supervised learning, sequence-based, and unsupervised learning approaches, the event management system 200 may generate accurate risk and readiness assessments that inform decision-making processes. The bounded propagation and scenario simulation capabilities may reduce computational overhead while maintaining accuracy, allowing organizations to evaluate multiple operational scenarios efficiently. This integrated approach may improve

operational efficiency by providing real-time insights into event readiness, enabling proactive identification of potential issues, and supporting rapid response to changing conditions.

[0065] Unless otherwise expressly indicated herein, all numerical values indicating mechanical/thermal properties, compositional percentages, dimensions and/or tolerances, or other characteristics are to be understood as modified by the word “about” or “approximately” in describing the scope of the present disclosure. This modification is desired for various reasons including industrial practice; material, manufacturing, and assembly tolerances; and testing capability.

[0066] As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0067] The description of the disclosure is merely exemplary in nature and, thus, variations that do not depart from the substance of the disclosure are intended to be within the scope of the disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure.

[0068] In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information, but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element

B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

[0069] In this application, the term controller or module may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality, such as, but not limited to, movement drivers and systems, transceivers, routers, input/output interface hardware, among others; or a combination of some or all of the above, such as in a system-on-chip.

[0070] The term memory is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

[0071] The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general-purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.

CLAIMS

What is claimed is:

1. A method for managing an event, the method comprising:
obtaining a dependency graph comprising a plurality of nodes representing operational elements of the event and a plurality of edges representing relationships between the operational elements;
generating, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine; and
performing a graph updating routine, wherein the graph updating routine comprises:
applying a modification to a first set of nodes of the plurality of nodes;
identifying a subgraph of the dependency graph based on the modification, wherein the subgraph comprises a second set of nodes of the plurality of nodes;
generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector; and
displaying the modified dependency graph.
2. The method of Claim 1, wherein the graph updating routine comprises a scenario simulation routine or a bounded propagation routine.
3. The method of Claim 1, wherein applying the modification further comprises modifying, for each node of the first set of nodes, at least one risk score of the risk vector.

4. The method of Claim 3, wherein the at least one risk score comprises a likelihood value, an impact value, a detection confidence value, and a data quality value.

5. The method of Claim 1, wherein the modification comprises modifying, for each node of the first set of nodes, at least one readiness score of the readiness vector.

6. The method of Claim 5, wherein the at least one readiness score comprises at least one of a completion status value, an inbound risk propagation values, a service level agreement (SLA) impact factor value, and a reliance value.

7. The method of Claim 1, wherein the analytic routine comprises at least one of a graph-based analytic routine, a statistical analytic routine, a supervised learning routine, a sequence-based routine, and an unsupervised learning routine.

8. The method of Claim 1, further comprising selectively adjusting, for a first node of the plurality nodes, at least one of the risk vector and the readiness vector when a relationship type associated with the first node is a Blocks type.

9. The method of Claim 1, wherein a number of nodes of the second set of nodes is based on at least one of a convergence threshold and a maximum iteration count, and wherein the method further comprises terminating the graph updating routine when (i) at least one readiness score of the readiness vector is less than the convergence

threshold or (ii) an iteration value of the graph updating routine is equal to the maximum iteration count.

10. A system for managing an event, the system comprising:

a processor; and

a nontransitory computer-readable medium comprising instructions that are executable by the processor, wherein the instructions comprise:

obtaining a dependency graph comprising a plurality of nodes representing operational elements of the event and a plurality of edges representing relationships between the operational elements;

generating, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine; and

performing a graph updating routine, wherein the graph updating routine comprises:

applying a modification to a first set of nodes of the plurality of nodes;

identifying a subgraph of the dependency graph based on the modification, wherein the subgraph comprises a second set of nodes of the plurality of nodes;

generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector; and

displaying the modified dependency graph.

11. The system of Claim 10, wherein the graph updating routine comprises a scenario simulation routine or a bounded propagation routine.

12. The system of Claim 10, wherein applying the modification further comprises modifying, for each node of the first set of nodes, at least one risk score of the risk vector.

13. The system of Claim 12, wherein the at least one risk score comprises a likelihood value, an impact value, a detection confidence value, and a data quality value.

14. The system of Claim 10, wherein the modification comprises modifying, for each node of the first set of nodes, at least one readiness score of the readiness vector.

15. The system of Claim 14, wherein the at least one readiness score comprises at least one of a completion status value, an inbound risk propagation values, a service level agreement (SLA) impact factor value, and a reliance value.

16. The system of Claim 10, wherein the analytic routine comprises at least one of a graph-based analytic routine, a statistical analytic routine, a supervised learning routine, a sequence-based routine, and an unsupervised learning routine.

17. The system of Claim 10, further comprising selectively adjusting, for a first node of the plurality nodes, at least one of the risk vector and the readiness vector when a relationship type associated with the first node is a Blocks type.

18. The system of Claim 10, wherein a number of nodes of the second set of nodes is based on at least one of a convergence threshold and a maximum iteration count, and wherein the instructions further comprise terminating the graph updating routine when (i) at least one readiness score of the readiness vector is less than the convergence threshold or (ii) an iteration value of the graph updating routine is equal to the maximum iteration count.

19. A system for managing an event, the system comprising:
a processor; and
a nontransitory computer-readable medium comprising instructions that are executable by the processor, wherein the instructions comprise:
generating a dependency graph comprising a plurality of nodes representing operational elements of the event and a plurality of edges representing relationships between the operational elements;
generating, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine, wherein the analytic routine comprises at least one of a graph-based analytic routine, a statistical analytic routine, a supervised learning routine, a sequence-based routine, and an unsupervised learning routine; and

performing a graph updating routine, wherein the graph updating routine comprises a scenario simulation routine or a bounded propagation routine, and wherein the graph updating routine comprises:

applying a modification to a first set of nodes of the plurality of nodes;

identifying a subgraph of the dependency graph based on the modification, wherein the subgraph comprises a second set of nodes of the plurality of nodes;

generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector; and

displaying the modified dependency graph.

20. The system of Claim 19, wherein applying the modification further comprises modifying, for each node of the first set of nodes, at least one risk score of the risk vector and at least one readiness score of the readiness vector.

ABSTRACT

A method for managing an event includes: obtaining a dependency graph including a plurality of nodes representing operational elements of the event and a plurality of edges representing relationships between the operational elements; generating, for each of the plurality of nodes, a risk vector and a readiness vector based on an analytic routine; and performing a graph updating routine, wherein the graph updating routine includes: applying a modification to a first set of nodes of the plurality of nodes; identifying a subgraph of the dependency graph based on the modification, wherein the subgraph includes a second set of nodes of the plurality of nodes; generating a modified dependency graph by updating, for each node of the second set of nodes, the risk vector and the readiness vector; and displaying the modified dependency graph.

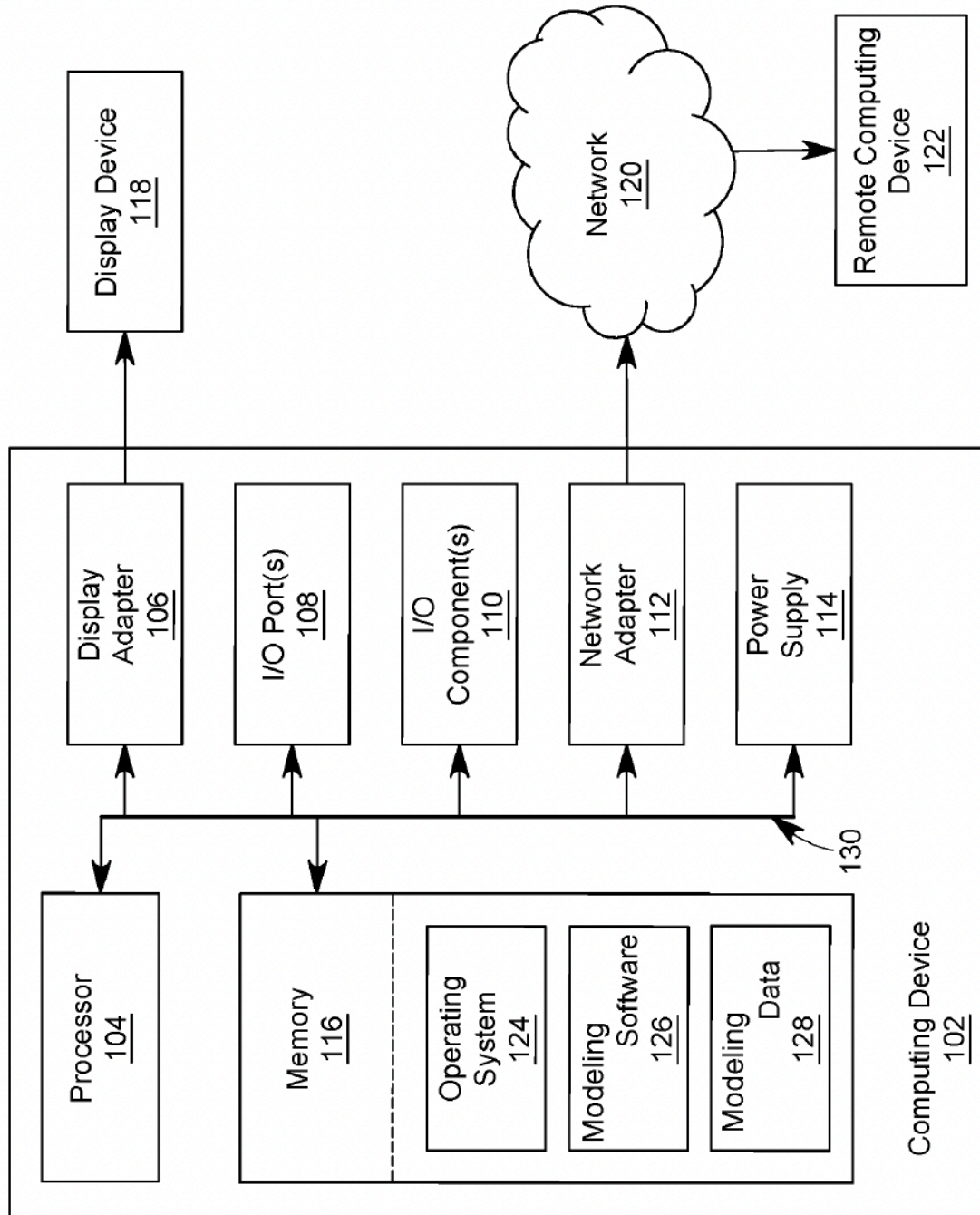


FIG. 1

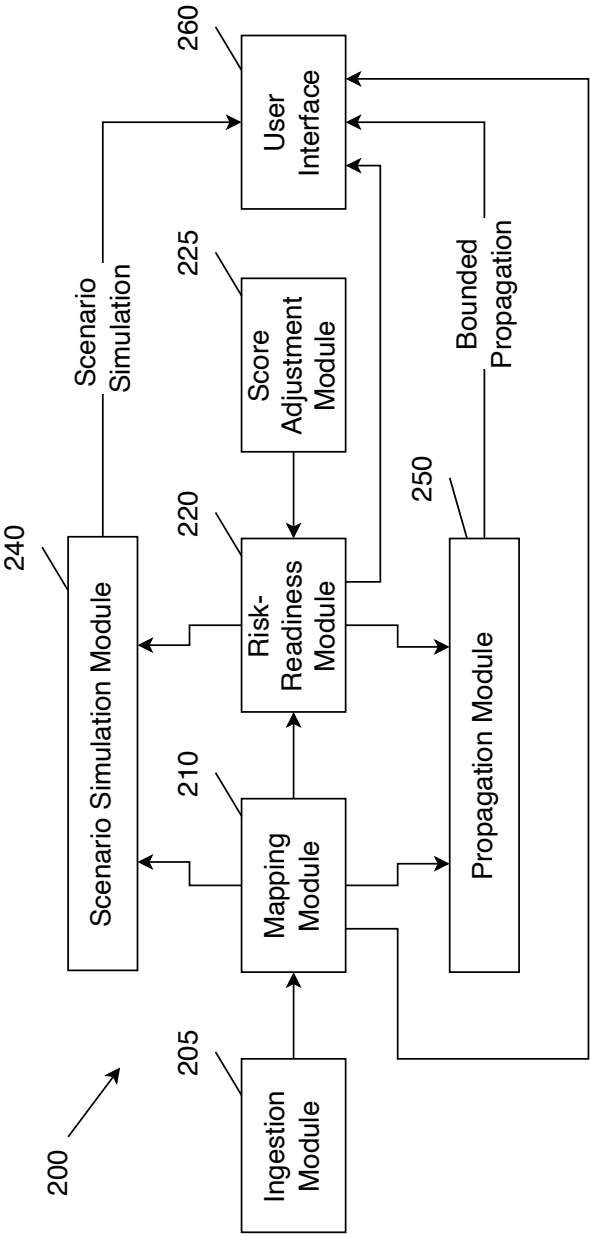


FIG. 2

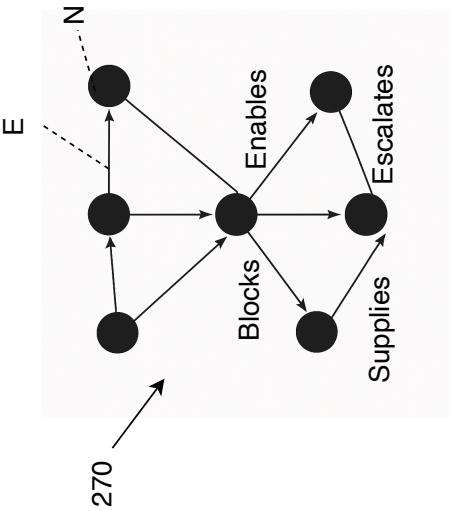


FIG. 3

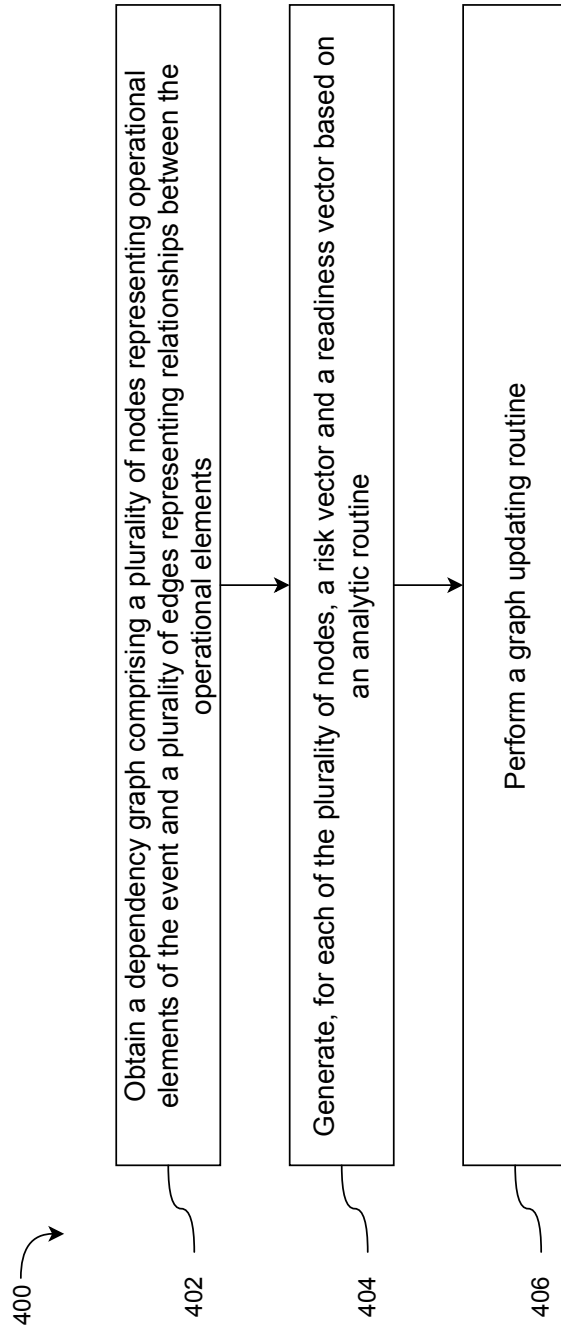


FIG. 4

**ASSIGNMENT AND DECLARATION (37 C.F.R. 1.63) FOR UTILITY, PROVISIONAL OR
DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 C.F.R. 1.76)**

Invention Title	SYSTEMS AND METHODS FOR MANAGING AN EVENT
This declaration is directed to: The attached application, or United States application or PCT international application number _____ filed on _____. (I hereby authorize the insertion of the application filing date and number into this form when they become known.) The above-identified application was made or authorized to be made by me. I hereby state that I have reviewed and understand the contents of the specification in the above-identified application, including the claims. I believe that I am the original inventor or an original joint inventor of a claimed invention in the application. I acknowledge my duty to disclose information which is material to patentability as defined in 37 C.F.R. 1.56, including for continuation-in-part applications, material information which became available between the filing date of the prior application and the national or PCT international filing date of the continuation-in- part application. Pursuant to my employment agreement heretofore made or by contractual obligation, I do sell, assign and set over unto Resilient Reach IP Holdings, LLC, a Delaware Company, having a place of business at 3651 Peachtree Pkwy, STE E-339, Suwanee, GA 30024 hereinafter referred to as the ASSIGNEE, its successors, assigns or other legal representatives, the entire right, title and interest, domestic and foreign, in and to the inventions and discoveries in the above-identified invention including the right of the ASSIGNEE, its successors, assigns or other legal representatives to make applications and to receive Letters Patent for the inventions and discoveries in any and all countries, entities and intergovernmental organizations, in its or their own name or names, or in my name, at its or their election (and I hereby authorize the ASSIGNEE or ASSIGNEE's attorney to insert the serial/application number and filing date of the application to receive Letters Patent herein: Serial No. _____, Filing Date _____), and I hereby sell, assign and set over to the ASSIGNEE, its successors, assigns or other legal representatives, all rights of priority, including any provisional applications, in and to the inventions and discoveries in any and all countries, entities and intergovernmental organizations including all applications claiming benefit of the filing date hereof, continuations, continuations-in-part, divisionals, reexaminations and reissue applications. And I hereby agree for myself, my heirs, successors, assigns or other legal representatives to execute any and all papers, including applications for Letters Patent of any and all kinds and in any and all countries, entities and intergovernmental organizations, and to perform any and all acts which the ASSIGNEE, its successors, assigns or other legal representatives may deem necessary to secure thereto the rights herein sold, assigned and set over. And I hereby represent and warrant that I have not granted any rights inconsistent with the rights granted herein. I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years or both.	
LEGAL NAME OF INVENTOR Inventor: <u>Tiwanne Hayes</u> Date: <u>11/10/2025</u>. Signature: <u>/Tiwanne Hayes/</u>	
Note: An Application Data Sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional declaration form for each additional inventor.	